Origin of Life and Self-Replication

PAX L5 Narrow L2 General 27B | Anticloud FZ LLE

Proposal Title: The Stochastic Autocatalytic Loop (SAL): A Kinetic Bridge from Prebiotic Chemistry to Darwinian Evolution

1. Core Hypothesis

The origin of self-replication did not begin with a single, complex "master molecule" (e.g., the RNA World's specific ribozyme) nor purely random assembly. Instead, it emerged through **Stochastic Autocatalytic Loops (SALs)** within spatially constrained micro-environments (e.g., hydrothermal pore networks or evaporating tidal pools).

In this framework, life is defined as a transition in a chemical system where the **information-to-mass ratio** becomes positive due to the emergence of a closed catalytic cycle that outcompetes background thermal noise. Specifically, we posit that short oligomers ($N < 20$ monomers) formed stochastic catalytic interactions where Product A accelerates its own formation from precursors $P_1...P_k$, creating a non-linear kinetic feedback loop. When these loops become coupled (hypercycles) and encapsulated by amphiphilous boundaries, they cross the threshold for selection.

Key Distinction: Unlike traditional "RNA-first" models requiring high-fidelity replication immediately, SAL allows for **low-fidelity, error-prone autocatalysis** initially. Selection acts on the *kinetic stability* of the network topology rather than the sequence fidelity of individual strands until a critical complexity threshold is reached.

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2. Mathematical Framework

We model the prebiotic soup as a set of interacting species $x_i(t)$ representing concentrations of molecular templates. The dynamics are governed by a generalized replicator equation augmented with synthesis rates, degradation, and stochastic fluctuations.

2.1 The Kinetic Master Equation

For a population of M distinct molecular types in a volume V:

$$ \frac{dx_i}{dt} = k_{syn,i}(x) - k_{deg,i} x_i + \eta_i(t) $$

Where:

* $k_{syn,i}(x) = \sum_{j=1}^{M} \alpha_{ij} x_j^{\beta} f(x_j, \text{template}_i)$ represents autocatalytic production. Here, α_{ij} is the catalytic efficiency matrix, and β is the reaction order (typically $\beta > 1$ for autocatalysis, implying super-linear growth).
* $k_{deg,i}$ accounts for hydrolysis, UV damage, and dilution.
* $\eta_i(t)$ represents intrinsic stochastic noise (fluctuations in collision frequency), modeled as Gaussian white noise scaled by $\sqrt{x_i}$.

2.2 The Eigen-Quasispecies Transition

To analyze the emergence of dominant sequences, we utilize the Jacobian matrix J of the system at equilibrium. The condition for a stable autocatalytic state (the "Eigenvalue Threshold") requires:

$$ \lambda_{max}(J) > \delta_{avg} $$

Where λ_{max} is the largest eigenvalue of the interaction matrix $\mathbf{A}$ (where $A_{ij} \propto \alpha_{ij}$) and δ_{avg} is the average decay/dilution rate.

If $\lambda_{max} \leq \delta_{avg}$, the system dissolves into a "death valley" of random polymers. If $\lambda_{max} > \delta_{avg}$, an exponential amplification occurs, selecting for the eigenvector corresponding to λ_{max}. This eigenvector represents the "winning" chemical composition.

2.3 Error Catastrophe Boundary

Let μ be the per-base mutation/error rate during template copying. For a genome of length L, the information content is preserved only if:

$$ \mu L < \ln(2) $$

However, in the early SAL phase, we hypothesize a mechanism where **combinatorial coupling** reduces the effective L required for viability. We define an "effective information capacity" I_{eff}:

$$ I_{eff} = L \cdot (1 - H(\mu)) $$

Where $H(\mu)$ is the binary entropy function. The system transitions to evolution when I_{eff} exceeds the minimum bits required to encode a metabolic closure.

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3. Proof Sketch / Key Steps

Step 1: Demonstration of Non-Linearity in Simple Oligomers

* *Premise:* Short RNA/DNA analogues (or peptide nucleic acids) exhibit catalytic activity even without folding into complex tertiary structures, provided local concentration is high.
* *Derivation:* Show that in a confined volume V_c (micropore), the probability of two complementary strands meeting scales as $O(1/V_c^2)$, while degradation scales linearly with time. There exists a critical volume V_{crit} below which autocatalytic terms dominate degradation terms, satisfying $\lambda_{max} > \delta_{avg}$.

Step 2: Emergence of Hypercycles via Cross-Catalysis

* *Logic:* Single autocatalysts are vulnerable to parasitic sequences (molecules that replicate but do not contribute to the loop).
* *Mechanism:* Introduce a second species Y such that X catalyzes Y and Y catalyzes X (mutualism).
* *Mathematical Result:* Solve the coupled differential equations for the (X, Y) system. Prove that the joint eigenvalue $\lambda_{XY} = \sqrt{k_{xy} k